



AgriTECH: Can India Lead The Global Race?

Given the rise in the number of agritech focused startups taking shape in the country and reforms being introduced by the government of India, can India lead the global race in agritech?

Mukul Yudhveer Singh

Refer to any report with keywords 'Top 10 Agriculture Nations of the World,' and India's name will pop up there. In terms of technical expertise, on the other hand, India has made significant advancements during the last two decades. Now, given the rise in the number of agritech focused startups taking shape in the country and reforms being introduced by the government of India, we are trying to answer one question—Can India lead the global race in agritech?

India: The advantages and disadvantages

Among the nations in 'Top 10 Agriculture Countries of the World' list, the United States, China, Russia, Brazil, and India grab the top spots. Compare these countries with India in terms of technological prowess in agriculture, and you will get your answer, but there is a catch.

If we compare India's agritech vertical with that of Europe or some other countries, it is on par with them. However, the advantage that those countries have is the bigger land sizes available for cultivation compared to India. It is easier to get ROI over larger pieces of land.

The Indian agritech entrepreneurs may not have the advantage of larger land size

and economies of scale, but the market size in India is pretty big. The real challenge for us is to develop and deploy technology in the local context.

Giving examples of 30-to 50-tonne capacity cold rooms in countries like Kenya and some from Europe, Devendra Gupta from Ecozen points out that such cold rooms do not work in India. The primary reason behind the same is smaller patches of lands owned by farmers, and hence smaller yields produced. He suggests making these cold rooms less technology-heavy and more efficiency heavy. Putting in more technology to a cold room might increase the cost to a farmer, but it is not necessary that the add-on technologies will increase the efficiency for the farmers.

Smallholder farmers, according to the Food and Agriculture Organisation (FAO) of the United Nations, are vital for India's agriculture and rural economy. Smallholder farmers—defined as those marginal and sub-marginal farm households that own or/and cultivate less than 2.0 hectares of land—constitute about 78 per cent of the country's farmers (as per Agricultural Census 1990-91). In absolute numbers, smallholder families constitute more than half of the national population of India.

"The tech that we are building in India